

Introductory Concepts of Probability

Formula Reference Sheet & Graded Problem Set with Full Solutions · Dr. Adebayo F. A.

- Chapter 1 — Set Theory ● Chapter 2 — Probability ● Chapter 3 — Conditional Probability & Bayes
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CHAPTER 01 Set Theory

DEFINITION 1.1.1

Set

A **set** is a collection of distinguishable objects or entities. Sets are denoted by uppercase letters $A, B, C, \dots$. Some reserved symbols:

- $\mathbb{N} = \{0, 1, 2, \dots\}$ — natural numbers
- $\mathbb{Z} = \{0, \pm 1, \pm 2, \dots\}$ — integers
- $\mathbb{R}$ — real numbers

A set must be **well-defined** (for any object, membership is decidable) and elements are **distinct** (no repetition).

Two ways to define a set:

- *Roster/Listing:* $A = \{2, 16, 9\}$
- *Set-builder rule:* $B = \{x \mid -2 \leq x \leq 2\}$

DEFINITION 1.1.2

Universal Set / Sample Space

U or S — the set of all objects under consideration

In mathematics: universal set U . In statistics: sample space S . Every set discussed is a subset of S .

DEFINITION 1.1.3

Empty Set

$\emptyset$ — the set containing no elements

Also called the null set. $\emptyset \subset A$ for every set A . In probability, $\emptyset$ is the impossible event.

DEFINITION 1.1.4

Subset

$A \subset B$ iff every element of A is also in B

Equivalently: $B \supset A$ reads " B contains A "

$A \not\subset B$ means at least one element of A is not in B .

DEFINITION 1.1.5 & 1.1.6

Equal Sets & Power Set

$A = B \Leftrightarrow A \subset B$ and $B \subset A$

$\mathcal{P}(A)$ = set of all subsets of A ; $|\mathcal{P}(A)| = 2^{|A|}$

e.g.

$A = \{1, 2, 3\} \Rightarrow \mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, A\}$
 — 8 subsets = 2^3 .

KEY DISTINCTION — $\subset$ VS $\in$

$\in$ relates an *element* to a set: $3 \in \{1, 2, 3\}$. $\subset$ relates a *set* to a set: $\{2, 3\} \subset \{1, 2, 3\}$. Never write $\{2, 3\} \in \{1, 2, 3\}$ — a set is not an element of a plain list set unless explicitly listed as such. Also: $\emptyset \subset A$ always (vacuously true), but $\emptyset \in A$ only if $\emptyset$ is explicitly an element of A .

DEFINITION 1.2.1

Intersection

$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

For n sets: $\bigcap_{i=1}^n A_i$

Disjoint: $A \cap B = \emptyset$ — no elements in common. Also called *mutually exclusive in probability*.

DEFINITION 1.2.3

Union

$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

For n sets: $\bigcup_{i=1}^n A_i$

"or" is *inclusive* — elements in both A and B are included.
 Union never shrinks a set: $A \subset A \cup B$.

DEFINITION 1.2.4

Complement

$\bar{A} = A^c = \{x \mid x \in S, x \notin A\}$

$A \cup \bar{A} = S \quad A \cap \bar{A} = \emptyset$

$\bar{\bar{S}} = \emptyset$ and $\bar{\emptyset} = S$. Double complement: $\bar{\bar{A}} = A$.

DEFINITION 1.2.5

Set Difference

$A - B = \{x \mid x \in A, x \notin B\} = A \cap \bar{B}$

$\bar{A} = S - A$. Set difference is not commutative:

$A - B \neq B - A$ in general. $A - B = A$ iff $A \cap B = \emptyset$.

EDGE CASE — STRICT INEQUALITIES IN SET-BUILDER NOTATION

When finding $P \cap Q$ where $P = \{x \mid 0 \leq x \leq 10\}$ and $Q = \{x \mid 5 < x < 20\}$, the endpoint $x = 5$ is in P but **not** in Q (strict inequality). So $P \cap Q = \{x \mid 5 < x \leq 10\}$. Always pay attention to $<$ vs $\leq$ at boundaries.

DEFINITION 1.3.1

Venn Diagram

A **Venn diagram** is a pictorial representation of sets. The sample space S is drawn as a large rectangle; sets are represented as circles or other shapes within it. Shaded regions represent the set operation of interest.

Common regions to recognise:

- Overlapping circles = non-disjoint sets ($A \cap B \neq \emptyset$)
- Non-overlapping circles = disjoint sets ($A \cap B = \emptyset$)
- One circle entirely inside another = subset ($A \subset B$)
- Region inside rectangle but outside all circles = complement region

THEOREM 1.4.1 – DISTRIBUTIVE LAWS (WITH PROOF)

Distributive Laws

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Proof sketch (first law): Let $x \in A \cap (B \cup C)$. Then $x \in A$ and $x \in B \cup C$, so $x \in A$ and ($x \in B$ or $x \in C$). This gives $x \in (A \cap B) \cup (A \cap C)$, i.e. $x \in (A \cap B) \cup (A \cap C)$. The reverse inclusion is symmetric. $\square$

THEOREM 1.4.3 – DE MORGAN'S LAWS

De Morgan's Laws

$$\overline{A \cup B} = \bar{A} \cap \bar{B} \quad \overline{A \cap B} = \bar{A} \cup \bar{B}$$

$$\text{Generalised: } \overline{\bigcup_{i=1}^n A_i} = \bigcap_{i=1}^n \bar{A}_i \text{ and } \overline{\bigcap_{i=1}^n A_i} = \bigcup_{i=1}^n \bar{A}_i$$

Verbal: "Not (A or B)" = "Not A and Not B ". "Not (A and B)" = "Not A or Not B ". These are the most commonly used identities for simplifying probability expressions.

WORKED EXAMPLES – ALGEBRA OF SETS

Key Proof Techniques

Example 1.4.2: Show $(A - B) - C = A - (B \cup C)$

$$(A - B) - C = (A \cap \bar{B}) \cap \bar{C} = A \cap (\bar{B} \cap \bar{C}) = A \cap \overline{(B \cup C)} = A - (B \cup C). \quad \square$$

Example 1.4.3: Show $A - (B \cup C) = (A - B) \cap (A - C)$

$$A - (B \cup C) = A \cap \overline{(B \cup C)} = A \cap (\bar{B} \cap \bar{C}) = (A \cap \bar{B}) \cap (A \cap \bar{C}) = (A - B) \cap (A - C). \quad \square$$

Example 1.4.1: Show $A \subset B \Leftrightarrow A = A \cap B$

($\Rightarrow$) If $A \subset B$, then $A \cap A \subset A \cap B$, but $A \cap B \subset A$ always, so $A = A \cap B$. ($\Leftarrow$) If $A = A \cap B \subset B$. $\square$

Statistics

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PROPERTY	STATEMENT	NOTES
Commutativity	$A \cup B = B \cup A, A \cap B = B \cap A$	Order doesn't matter
Associativity	$(A \cup B) \cup C = A \cup (B \cup C)$	Grouping doesn't matter
Identity	$A \cup \emptyset = A, A \cap S = A$	$\emptyset$ is identity for $\cup$; S is identity for $\cap$
Annihilation	$A \cap \emptyset = \emptyset, A \cup S = S$	S absorbs under $\cup$
Idempotency	$A \cup A = A \cap A = A$	Repeating an operation on itself is harmless
Complement laws	$A \cup \bar{A} = S, A \cap \bar{A} = \emptyset, \bar{\bar{A}} = A$	Double negation cancels
Monotone subset	$A \cap B \subset A \subset A \cup B$	Intersection shrinks; union grows
Difference identity	$A - B = A \cap \bar{B}$	Used to convert difference into intersection
Useful chain	$(A - B) - C = A - (B \cup C)$	Proof: De Morgan + associativity
Another chain	$A - (B \cup C) = (A - B) \cap (A - C)$	Distributing difference over union

COMMON MISTAKE – DE MORGAN DIRECTION

Students often flip De Morgan's laws. Remember: the **bar goes over the whole expression** and the **operation flips** ($\cup \leftrightarrow \cap$). So $\overline{A \cup B} = \bar{A} \cap \bar{B}$ (NOT $\bar{A} \cup \bar{B}$). A memory trick: "Break the bar, change the operator." Also note: $\overline{A \cup B} \neq \bar{A} \cup \bar{B}$ — the latter equals $\overline{A \cap B}$.

CHAPTER 02

Probability Concepts

DEFINITION

Sample Space & Events

S = set of all possible outcomes

Event: any subset $A \subseteq S$. Elementary event: $|\{a\}| = 1$.

Mutually exclusive: $A \cap B = \emptyset$.

FREQUENCY INTERPRETATION

Empirical Probability

$$P(A) = \lim_{n \rightarrow \infty} \frac{r}{n}$$

r = number of times A occurs in n trials.

AXIOMS (KOLMOGOROV)

Axioms of Probability**AI.** $0 \leq P(A) \leq 1$ for all $A \subseteq S$ **AII.** $P(S) = 1$ **AIII.** $A \cap B = \emptyset \Rightarrow P(A \cup B) = P(A) + P(B)$ General: $P(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i)$ when pairwise mutually exclusive

THEOREM 2.2.1

Complement Rule

$$P(\bar{A}) = 1 - P(A)$$

THEOREM 2.2.2

Partition Rule

$$P(A) = P(A \cap B) + P(A \cap \bar{B})$$

Follows from $A = (A \cap B) \cup (A \cap \bar{B})$, which are disjoint.

THEOREM 2.2.3

Addition Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

THEOREM 2.2.4

Monotonicity

$$A \subset B \Rightarrow P(A) \leq P(B)$$

EQUALLY LIKELY OUTCOMES

Classical Probability Formula

$$P(A) = \frac{n(A)}{N}$$

 $n(A)$ = number of favourable outcomes; $N = n(S)$ = total number of equally likely outcomes.

CHAPTER 2 – COUNTING

Permutations & CombinationsPermutations of n objects: $n!$ Permutations of n taken r at a time: ${}_nP_r = \frac{n!}{(n-r)!}$ Permutations with repetition: n^r Combinations: $\binom{n}{r} = \frac{n!}{(n-r)!r!}$ Multinomial coefficient: $\binom{n}{n_1, n_2, \dots, n_r} = \frac{n!}{n_1!n_2!\dots n_r!}$ $\sum_{i=1}^r n_i = n$. Counts ways to divide n distinct items into r groups of sizes $n_1, \dots, n_r$.

DEFINITION 3.1.1

Conditional Probability

$$P(B | A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) > 0$$

Satisfies all three probability axioms; $P(B | A) + P(\bar{B} | A) = 1$.

MULTIPLICATION RULE

Chain Rule

$$P(A_1 \cap A_2 \cap \dots \cap A_n) = P(A_1) P(A_2 | A_1) P(A_3 | A_2, A_1) \dots P(A_n | A_{n-1}, \dots, A_1)$$

DEFINITION 3.3.1

Independence

$$A \text{ and } B \text{ independent} \Leftrightarrow P(A \cap B) = P(A) P(B)$$

Equivalently $P(B | A) = P(B)$. If $A \perp B$ then also $A \perp \bar{B}$, $\bar{A} \perp B$, $\bar{A} \perp \bar{B}$.

THEOREM

Law of Total Probability

$$P(B) = \sum_{i=1}^n P(B | A_i) P(A_i)$$

Requires $\{A_i\}$ to be pairwise mutually exclusive and exhaustive: $\bigcup_i A_i = S$.

THEOREM 3.2.1

Bayes' Theorem (General Form)

$$P(A_k | B) = \frac{P(B | A_k) P(A_k)}{\sum_{i=1}^n P(B | A_i) P(A_i)}$$

$P(A_k)$ = prior; $P(B | A_k)$ = likelihood; $P(A_k | B)$ = posterior.

Bayes' Theorem (Two-Event Form)

$$P(A \mid B) = \frac{P(B \mid A) P(A)}{P(B \mid A) P(A) + P(B \mid \bar{A}) P(\bar{A})}$$

FORMULA	EXPRESSION	NOTES
Conditional probability	$P(B \mid A) = \frac{P(A \cap B)}{P(A)}$	—
Complement of conditional	$P(\bar{B} \mid A) = 1 - P(B \mid A)$	—
Partition of A	$P(A) = P(A \cap B) + P(A \cap \bar{B})$	—
Addition rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	—
Independence check	$P(A \cap B) \stackrel{?}{=} P(A)P(B)$	also check $P(A \cup B) \leq 1$
Total probability (2 parts)	$P(B) = P(B \mid A)P(A) + P(B \mid \bar{A})P(\bar{A})$	—



Common Exam Traps & Pitfalls

Mistakes that cost marks — know these before you sit the exam.

SET THEORY · TRAP 1



Confusing $\subset$ with $\in$

✗ $D \in C$ when $D = \{2, 4\}$ and $C = \{1, 2, 3, 4\}$

✓ $D \subset C$ — a *set* is a subset, not an element

$\in$ means "is an element of"; the element must be a single object, not a set (unless the set explicitly contains sets as members).

SET THEORY · TRAP 2



Mixing discrete and continuous sets

✗ $C = E$ because both involve 1 to 4

✓ $C = \{1, 2, 3, 4\}$ (integers) $\neq E = [1, 4]$ (all reals)

Always check whether a set is defined by listing integers or by an inequality over the reals before claiming equality.

PROBABILITY · TRAP 3

Forgetting to subtract the intersection

$$\times P(A \cup B) = P(A) + P(B)$$

$$\checkmark P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The simple sum double-counts outcomes in $A \cap B$. Only omit the subtraction when events are mutually exclusive ($P(A \cap B) = 0$).

PROBABILITY · TRAP 4

Checking impossibility constraints

$$\times \text{Accepting } P(A \cap B) > P(A) \text{ or } P(A \cup B) < \max(P(A), P(B))$$

$$\checkmark \text{Both violate monotonicity: } A \cap B \subseteq A \text{ forces } P(A \cap B) \leq P(A)$$

Before computing, verify:

$$P(A \cap B) \leq \min(P(A), P(B)) \text{ and}$$

$$P(A \cup B) \geq \max(P(A), P(B)).$$

CONDITIONAL · TRAP 5

Reversing $P(A | B)$ and $P(B | A)$

$$\times$$

$$P(\text{Machine A} | \text{defective}) = P(\text{defective} | \text{Machine A})$$

$$\checkmark \text{Apply Bayes' Theorem — the two are generally unequal}$$

This is the "prosecutor's fallacy." Always identify which event is given and which is being sought.

CONDITIONAL · TRAP 6

Independence $\neq$ Mutual Exclusivity

$$\times \text{If } P(A \cap B) = 0 \text{ then } A \text{ and } B \text{ are independent}$$

$$\checkmark \text{Mutual exclusivity means } P(A \cap B) = 0; \text{ independence requires } P(A \cap B) = P(A)P(B)$$

Two non-trivial mutually exclusive events are dependent (knowing one occurred rules out the other).

BAYES · TRAP 7

Ignoring base rates (prior probability)

$$\times \text{A 98\% accurate test} \implies \text{positive result means 98\% chance of disease}$$

$$\checkmark \text{With a rare disease (e.g. 3\% prevalence), most positives are false positives — use Bayes}$$

Low prior probability can make $P(\text{disease} | +)$ far smaller than the test's sensitivity. Always include $P(C)$ in the denominator via total probability.

COUNTING · TRAP 8

Order matters: permutation vs combination

$$\times \text{Using } \binom{n}{r} \text{ when the arrangement (order) is significant}$$

$$\checkmark \text{Use } {}_n P_r = n! / (n - r)! \text{ when order matters; } \binom{n}{r} \text{ when only selection matters}$$

Ask: "Does swapping two chosen items give a different outcome?" If yes, use permutations.

Forgetting zero-digit restrictions

✗ Treating each remaining digit position as having 9 choices after fixing the first

✓ If the second position also excludes 0 AND the first digit, count carefully: $10 - 2 = 8$ choices

Work position by position, accounting for all restrictions already applied.

Partition must be exhaustive

✗ Using

$$P(B) = P(B | A_1)P(A_1) + P(B | A_2)P(A_2)$$

when $A_1 \cup A_2 \neq S$

✓ Ensure $\{A_i\}$ covers all of S : $\sum_i P(A_i) = 1$

Missing partitions cause the computed total probability to be less than the true value.



Problem Sheets — Graded Questions

Questions drawn directly from the STA122 Problem Sheets. Difficulty is indicated for each item.

PROBLEM SHEET 1 — SET THEORY

Q1

EASY

Let $U = \{1, 2, 3, \dots, 10\}$. Three subsets are defined as $A = \{1, 2, 3, \dots, 8\}$, $B = \{2, 3, 4\}$ and $C = \{1, 3, 5, 7, 9\}$.

- Find $\bar{A}$, $B \cup \bar{C}$, and $A \cap (\bar{B} \cup C)$.
- Is it true that $\bar{B} \subset A \cap C$?

Q2

MEDIUM

Let $A = \{x \mid 0 < x < 5\}$, $B = \{0, 5\}$, $C = \{1, 2, 3, 4\}$, $D = \{2, 4\}$ and $E = \{x \mid 1 \leq x \leq 4\}$. Determine which statements are TRUE or FALSE:

- $C = E$
- $D \subset C$
- $D \in E$
- $E \not\subset A$

Q3

HARD

Show algebraically that $(A - B) - C = A - (B \cup C)$ using set identities.

Q4

MEDIUM

Let S be the set of all companies on the Botswana Stock Exchange and define: A = mining sector, B = turnover > P10m, C = financial year ending in June, D = share price higher than 6 months ago. Describe in words the set $B \cap (C \cup A)$.

PROBLEM SHEET 2 – PROBABILITY FOUNDATIONS

Q5

EASY

Let A and B be events in a sample space with $P(A) = 0.3$, $P(B) = 0.4$ and $P(A \cap B) = 0.2$. Find the probability of each of the following events:

1. Either A or B occurs.
2. A does not occur.
3. Neither A nor B occurs.
4. Exactly one of A or B occurs.

Q6

MEDIUM

Determine which of the following combinations are *impossible* (i.e. violate the axioms of probability):

1. $P(A) = 0.7$, $P(B) = 0.9$, $P(A \cap B) = 0.4$
2. $P(A) = 0.2$, $P(B) = 0.5$, $P(A \cap B) = 0.25$
3. $P(A) = 0.8$, $P(B) = 0.4$, $P(A \cup B) = 0.7$

Q7

HARD

Show that $P(A \cup \bar{B}) = 1 - [P(B) - P(A \cap B)]$.

Q8

MEDIUM

What is the probability that an 8-digit cellphone number has no repeated digits, given that the number must start with 7 and the second digit cannot be 0?

Q9

HARD

A pack of 52 cards is dealt into four bridge hands of 13 cards each. Find the probability that a given hand contains:

1. 3 spades, 4 diamonds, 1 heart and 5 clubs.
2. Exactly 3 aces and exactly 4 picture cards (J, Q, K).

Q10

MEDIUM

A class of 75 students is divided into tutorial groups of 25, 30 and 20.

1. In how many ways can this be done?
2. Two brothers are in the class. What is the probability they end up in the same tutorial group?

Q11

EASY

Suppose events A and B satisfy $P(A) = 0.6$, $P(B) = 0.2$, and $P(A | B) = 0.5$. Find:

1. $P(B | A)$
2. $P(A \cup B)$
3. $P(\bar{A} \cap B)$
4. $P(B | \bar{A})$

Q12

MEDIUM

Two machines produce the same item. Machine A produced 40% of last week's output; Machine B the remaining 60%. Defect rates are 10% for A and 4% for B.

1. What proportion of the entire output was defective?
2. Given that a randomly selected item is defective, what is the probability it came from Machine A?

Q13

HARD

An X-ray test detects a disease present in 3% of the population. The test falsely reports positive for 7% of healthy people and misses 2% of diseased people.

1. What proportion of diseased people are correctly diagnosed?
2. Of those with a positive reaction, what proportion actually have the disease?
3. Of those with a negative reaction, what proportion actually have the disease?
4. What proportion of tests give an incorrect diagnosis?

Q14

MEDIUM

If $P(A) = 0.6$, $P(B | A) = 0.3$ and $P(A \cup B) = 0.72$, determine whether A and B are independent, mutually exclusive, or neither.

Q15

HARD

A student passes Probability with probability 0.8 if she studied and 0.3 if she did not. 60% of the class studied. Given that a randomly chosen student passed, what is the probability she studied?

Answer Sheet — Full Solutions

Complete worked solutions for all fifteen questions above.

SOLUTIONS — PROBLEM SHEET 1

A1

1. $\bar{A} = \{9, 10\}$
2. $\bar{C} = \{2, 4, 6, 8, 10\}$, so $B \cup \bar{C} = \{2, 3, 4, 6, 8, 10\}$
3. $\bar{B} = \{1, 5, 6, 7, 8, 9, 10\}$, $\bar{B} \cup C = \{1, 3, 5, 6, 7, 8, 9, 10\}$, so $A \cap (\bar{B} \cup C) = \{1, 3, 5, 6, 7, 8\}$
4. Is $\bar{B} \subset A \cap C$? We have $\bar{B} = \{1, 5, 6, 7, 8, 9, 10\}$ and $A \cap C = \{1, 3, 5, 7\}$. Since $6 \in \bar{B}$ but $6 \notin A \cap C$, the statement is **FALSE**.

A2

1. $C = E$? $C = \{1, 2, 3, 4\}$ (integers) while $E = \{x : 1 \leq x \leq 4\}$ (all reals in $[1, 4]$). **FALSE**

2. $D \subset C$? $D = \{2, 4\} \subset \{1, 2, 3, 4\} = C$. **TRUE**

3. $D \in E$? D is a set, not a real number; it cannot be an element of E . **FALSE**

4. $E \not\subset A$? Every $x \in E = [1, 4]$ satisfies $0 < x < 5$, so $E \subset A$. Statement is **FALSE**.

A3

$$(A - B) - C = (A \cap \bar{B}) \cap \bar{C} = A \cap (\bar{B} \cap \bar{C}) = A \cap \overline{(B \cup C)} = A - (B \cup C). \quad \square$$

A4

$B \cap (C \cup A)$ is the set of companies whose annual turnover exceeds P10 million *and* whose financial year ends in June *or* which are in the mining sector (or both).

SOLUTIONS – PROBLEM SHEET 2

A5

1. $P(A \cup B) = 0.3 + 0.4 - 0.2 = 0.5$

2. $P(\bar{A}) = 1 - 0.3 = 0.7$

3. $P(\overline{A \cup B}) = 1 - 0.5 = 0.5$

4. $P(\text{exactly one}) = P(A \cup B) - P(A \cap B) = 0.5 - 0.2 = 0.3$

A6

1. $P(A \cap B) = 0.4 \leq \min(P(A), P(B)) = 0.7$. ✓ Possible.

2. $P(A \cap B) = 0.25 > P(A) = 0.2$. Since $A \cap B \subset A$, we need $P(A \cap B) \leq P(A)$. **Impossible.**

3. $P(A \cup B) = 0.7$. Check: must have $P(A \cup B) \geq \max(P(A), P(B)) = 0.8$. But $0.7 < 0.8$. **Impossible.**

A7

From the partition rule, $P(B) = P(B \cap A) + P(B \cap \bar{A})$, so $P(B \cap \bar{A}) = P(B) - P(A \cap B)$.

Then $P(A \cup \bar{B}) = 1 - P(\bar{A} \cap B) = 1 - [P(B) - P(A \cap B)]$. $\square$

SOLUTIONS – PROBLEM SHEET 3

A8

First digit is fixed as 7 (1 choice). Second digit: 0 is excluded and 7 already used, leaving $10 - 2 = 8$ choices.

Digits 3–8 are chosen from the remaining 8 digits without repetition.

Total (no restrictions beyond start=7, no-repeat): $1 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3$.

Favourable (second digit $\neq 0$): $1 \times 8 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3$.

$$P = \frac{1 \times 8 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3}{1 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3} = \frac{8}{9} \approx$$

0.8889

A9

1.

$$P = \frac{\binom{13}{3} \binom{13}{4} \binom{13}{1} \binom{13}{5}}{\binom{52}{13}}$$

$$= \frac{286 \times 715 \times 13 \times 1287}{\binom{52}{13}} \approx 0.00539$$

2. Aces: $\binom{4}{3}$. Picture cards (J,Q,K — 12 total): $\binom{12}{4}$. Remaining 6 from 36 others: $\binom{36}{6}$.

$$P = \frac{\binom{4}{3} \binom{12}{4} \binom{36}{6}}{\binom{52}{13}} \approx$$

0.00412

A10

$$1. \quad \binom{75}{25, 30, 20} = \frac{75!}{25! 30! 20!} \approx 5.99 \times 10^{35}$$

2. Fix both brothers in the same group — three cases:

- Both in group of 25: $\binom{73}{23, 30, 20}$
- Both in group of 30: $\binom{73}{25, 28, 20}$
- Both in group of 20: $\binom{73}{25, 30, 18}$

$$P = \frac{\binom{73}{23, 30, 20} + \binom{73}{25, 28, 20} + \binom{73}{25, 30, 18}}{\binom{75}{25, 30, 20}}$$

Simplifies via the multinomial identity to:

$$P = \frac{25 \cdot 24 + 30 \cdot 29 + 20 \cdot 19}{75 \cdot 74} = \frac{600 + 870 + 380}{5550} = \frac{1850}{5550} =$$

$$\frac{1}{3} \approx 0.3333$$

SOLUTIONS – PROBLEM SHEET 4

A11

$$1. P(A \cap B) = P(A | B) P(B) = 0.5 \times 0.2 = 0.10. \text{ Hence } P(B | A) = \frac{0.10}{0.60} = \frac{1}{6} \approx 0.167$$

$$2. P(A \cup B) = 0.6 + 0.2 - 0.1 = 0.7$$

$$3. P(\bar{A} \cap B) = P(B) - P(A \cap B) = 0.2 - 0.1 = 0.1$$

$$4. P(B | \bar{A}) = \frac{P(B \cap \bar{A})}{P(\bar{A})} = \frac{0.1}{0.4} = 0.25$$

A12

1. Let D =defective, M_A =from Machine A. By total probability:

$$P(D) = P(D | M_A)P(M_A) + P(D | M_B)P(M_B) = 0.10(0.4) + 0.04(0.6) = 0.04 + 0.024 =$$

$$0.064$$

2. By Bayes' theorem:

$$P(M_A | D) = \frac{0.10 \times 0.4}{0.064} = \frac{0.04}{0.064} =$$

$$0.625$$

A13

Let C =has disease (prior $P(C) = 0.03$), $+$ =positive test.

1. Sensitivity = $P(+ | C) = 1 - 0.02 = 0.98$

2. By Bayes: $P(C | +) = \frac{0.98 \times 0.03}{0.98(0.03) + 0.07(0.97)} = \frac{0.0294}{0.0294 + 0.0679} = \frac{0.0294}{0.0973} \approx 0.302$

3. $P(C | -) = \frac{P(- | C)P(C)}{P(-)} = \frac{0.02 \times 0.03}{0.02(0.03) + 0.93(0.97)} = \frac{0.0006}{0.0006 + 0.9021} \approx 0.000665$

4. Incorrect: false positive + false negative.

$$P(\text{error}) = P(+ | \bar{C})P(\bar{C}) + P(- | C)P(C) = 0.07(0.97) + 0.02(0.03) = 0.0679 + 0.0006 \approx 0.0685$$

A14

$$P(A \cap B) = P(B | A) P(A) = 0.3 \times 0.6 = 0.18.$$

Check addition rule: $P(A \cup B) = 0.6 + P(B) - 0.18 = 0.72 \Rightarrow P(B) = 0.30$.

Independence test: $P(A)P(B) = 0.6 \times 0.30 = 0.18 = P(A \cap B)$. **A and B are independent.**

(They are not mutually exclusive since $P(A \cap B) = 0.18 \neq 0$.)

A15

Let St =studied, Pa =passed. Given: $P(Pa | St) = 0.8$, $P(Pa | \bar{St}) = 0.3$, $P(St) = 0.6$.

Total probability: $P(Pa) = 0.8(0.6) + 0.3(0.4) = 0.48 + 0.12 = 0.60$.

Bayes: $P(St | Pa) = \frac{0.8 \times 0.6}{0.60} = \frac{0.48}{0.60} = 0.8$

CHAPTER 04 Random Variables

DEFINITION 4.1.1

Random Variable

A **random variable** is a function $X : S \rightarrow \mathbb{R}$ whose domain is a sample space S and whose range is the real line. It assigns a numerical value to every elementary outcome in S .

- **Random** — its value depends on the outcome of a random experiment.
- **Variable** — it takes on several possible values.

We use uppercase letters (X, Y, Z) for random variables and lowercase (x, y, z) for specific values they take.

COIN-FLIP EXAMPLE

Assigning Numerical Values

Toss 3 coins; X = number of heads.

$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$

$X : \quad 3, 2, 2, 2, 1, 1, 1, 0$

$P(X = 1) = 3/8$ since three outcomes map to $X = 1$.

KEY POINT

Event Language in Terms of X

$\{X = x\}$ means the set of outcomes

$\omega \in S$ with $X(\omega) = x$

$\{X \leq a\} = \{\omega : X(\omega) \leq a\}$

Probabilities are assigned to these events, not directly to values of X .

CLASSIFICATION

Discrete vs Continuous Random Variables

Discrete: takes isolated (countable) values — usually integers.

Continuous: can take any value in an interval on $\mathbb{R}$; between any two values there is another possible value.

Examples discrete: number of customers, number of defectives, number of goals. Examples continuous: waiting time, proportion, volume, distance.

EDGE CASE — CLASSIFICATION

Not every integer-valued variable is discrete in the strict sense — the key criterion is whether the range is countable (finite or countably infinite). Money amounts with cents are technically discrete but are often modelled as continuous. A variable like "last digit of a phone number" takes $\{0, 1, \dots, 9\}$ — discrete with a finite range. Beware: *the number of letters in the Greek alphabet is a constant (24), not a random variable at all.*

DEFINITION 4.3.1 — DISCRETE

Probability Mass Function (p.m.f.)

For a discrete random variable X , the **probability mass function** $p(x) = P(X = x)$ must satisfy:

1. $p(x) \geq 0$ for all x ; $p(x) = 0$ for all x not in the support.
2. $0 \leq p(x) \leq 1$ for all x .
3. $\sum_{\text{all } x} p(x) = 1$.

DISCRETE PROBABILITY

BAR GRAPH

Probability on an Interval

$$P(a \leq X \leq b) = \sum_{x=a}^b p(x)$$

Watch strict vs non-strict inequalities!

$$P(X > 3) \neq P(X \geq 3).$$

Visualising a p.m.f.

Plot $p(x)$ vs x as vertical bars.

Bar heights must be non-negative and sum to 1. Bars at non-integer or isolated points are fine — the p.m.f. need not be integer-valued.

EDGE CASE — STRICT VS NON-STRICT INEQUALITIES

For a **discrete** RV: $P(X \geq 3) = P(X = 3) + P(X = 4) + \dots$ while $P(X > 3) = P(X = 4) + P(X = 5) + \dots$. These differ by exactly $p(3)$. For a **continuous** RV, $P(X = c) = 0$ for any fixed c , so $P(X \geq 3) = P(X > 3)$. Never confuse the two types!

DEFINITION 4.3.2

Cumulative Distribution Function (c.d.f.)

For any random variable X , the c.d.f. is $F(a) = P(X \leq a)$.

For discrete X : $F(a) = \sum_{x \leq a} p(x)$. It is a **step function** — constant between jump points, jumping by $p(x_i)$ at each mass point x_i .

PROPERTIES OF THE C.D.F.

Four Key Properties

- (a) F is non-decreasing: $a < b \Rightarrow F(a) \leq F(b)$.
- (b) $\lim_{x \rightarrow \infty} F(x) = 1$.
- (c) $\lim_{x \rightarrow -\infty} F(x) = 0$.
- (d) F is right-continuous: $\lim_{a_n \downarrow a} F(a_n) = F(a)$.

READING THE CDF

Recovering Probabilities from F

$$P(X = a) = F(a) - F(a^-) \text{ (jump size at } a\text{)}$$

$$P(a < X \leq b) = F(b) - F(a)$$

$$P(X < a) = F(a^-) \text{ (value just before } a\text{)}$$

$$P(X > a) = 1 - F(a)$$

STEP FUNCTION EXAMPLE

CDF from a p.m.f.

If

$$p(1) = 1/4, p(2) = 1/2, p(3) = 1/8, p(4) = 1/8:$$

$$F(x) = 0 \text{ for } x < 1; 1/4 \text{ for } 1 \leq x < 2; 3/4 \text{ for } 2 \leq x < 3; 7/8 \text{ for } 3 \leq x < 4; 1 \text{ for } x \geq 4.$$

The function jumps at the mass points and is flat between them.

EDGE CASE – CDF AT JUMP POINTS

$P(X < 3)$ vs $P(X \leq 3)$: For a discrete RV, $P(X < 3) = F(2)$ (the value just before 3) while $P(X \leq 3) = F(3)$. These are different by $p(3)$. When reading a CDF table, always check whether the inequality includes the boundary.

DEFINITION 4.3.3 – CONTINUOUS

Probability Density Function (p.d.f.)

A function $f(x)$ is a p.d.f. if:

1. $f(x) \geq 0$ for all x (non-negative).
2. $\int_{-\infty}^{\infty} f(x) dx = 1$ (total area = 1).

For continuous X : $P(a \leq X \leq b) = \int_a^b f(x) dx$.

Critical: $P(X = c) = 0$ for any fixed c — individual points have zero probability.

CONTINUOUS CDF

CDF from the p.d.f.

$$F(a) = P(X \leq a) = \int_{-\infty}^a f(x) dx$$

For continuous RVs: $P(X < a) = P(X \leq a) = F(a)$ — the strict/non-strict distinction vanishes.

FINDING K

Normalisation Condition

To find unknown constant k : set $\int_{-\infty}^{\infty} f(x) dx = 1$ and solve for k .

Always integrate over the full support (where $f(x) \neq 0$), not over all of $\mathbb{R}$ unless the support is all of $\mathbb{R}$.

EDGE CASE – P.D.F. VALUES CAN EXCEED 1

Unlike a p.m.f., a p.d.f. value $f(x)$ is **not** a probability and can be greater than 1. For example $f(x) = 2x$ on $[0, 1]$ gives $f(0.9) = 1.8 > 1$. What must equal 1 is the integral (area under the curve), not individual function values.

§4.4.1 – EXPECTED VALUE

Expectation (Mean)

$$\text{Discrete: } E(X) = \sum_{x: p(x) > 0} x p(x)$$

$$\text{Continuous: } E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

The expected value is the probability-weighted average of all possible values. It need not be a value that X can actually take.

EXPECTATION OF A FUNCTION

Law of the Unconscious Statistician

$$\text{Discrete: } E[g(X)] = \sum_x g(x) p(x)$$

$$\text{Continuous: } E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$$

Useful for $g(X) = X^2$, $g(X) = aX + b$, $g(X) = (X - \mu)^2$, etc.

PROPOSITION 4.4.1

Linearity of Expectation

$$E(aX + b) = aE(X) + b$$

Holds for both discrete and continuous RVs. Constants pass through the expectation operator.

SPECIAL CASES

Useful Identities

$$E(c) = c \text{ for any constant } c$$

$$E(X - \mu) = 0 \text{ where } \mu = E(X)$$

$$E(X^2) = V(X) + [E(X)]^2$$

§4.4.2 – VARIANCE AND STANDARD DEVIATION

Variance

$$V(X) = E[(X - \mu)^2] = E(X^2) - \mu^2 \text{ where } \mu = E(X)$$

$$\text{Discrete: } V(X) = \sum_x (x - \mu)^2 p(x)$$

$$\text{Continuous: } V(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

Standard deviation: $SD(X) = \sqrt{V(X)}$. Variance of a constant is 0. $V(aX + b) = a^2 V(X)$.

EDGE CASE – SAME MEAN, DIFFERENT VARIANCE

Consider $W = 0$ always, $Y = \pm 1$ each with prob $1/2$, $Z = \pm 100$ each with prob $1/2$. All have $E = 0$, but $V(W) = 0$, $V(Y) = 1$, $V(Z) = 10000$. Variance captures *spread* — never infer spread from the mean alone.

QUANTITY	DISCRETE FORMULA	CONTINUOUS FORMULA
$E(X)$	$\sum x p(x)$	$\int x f(x) dx$
$E[g(X)]$	$\sum g(x) p(x)$	$\int g(x) f(x) dx$
$V(X)$	$E(X^2) - [E(X)]^2$	$E(X^2) - [E(X)]^2$
$E(aX + b)$	$aE(X) + b$	
$V(aX + b)$	$a^2V(X)$	



Problem Sheets 5 & 7 — Random Variables

Questions from Problem Sheets 5 and 7. Work through each before checking the solutions.

PROBLEM SHEET 5 — DEFINING RANDOM VARIABLES

Q16

EASY

State whether each of the following is a random variable; if so, classify it as discrete or continuous and write down the set of possible values.

1. The time required to answer this question.
2. The number of words in any randomly chosen book from the library.
3. The maximum temperature recorded at Sir Seretse Khama International Airport.
4. The number of application letters before successfully landing a job.
5. The proportion of students passing STA122.

Q17

MEDIUM

Suppose a die is rolled twice. Determine the possible values and their corresponding probabilities for each of the following random variables:

1. The maximum value appearing in both rolls.
2. The sum of the values from the two rolls.
3. The value of the first roll minus the value of the second roll.

Q18

MEDIUM

Two balls are chosen at random from an urn containing 6 white, 2 black and 2 red balls. You win P20.00 for each black ball selected and lose P10.00 for each white ball selected (red balls have no effect). Let X be your total winnings. What are the possible values of X and their probabilities?

Q19

HARD

A small pool building company is equally likely to complete 2 or 3 contracts each month. The company receives between 1 and 4 contracts per month with probabilities $P(1) = 0.1$, $P(2) = 0.2$, $P(3) = 0.5$, $P(4) = 0.2$. At the start of this month the company has 2 contracts carried forward. Let X be the number of contracts carried forward to next month. Assuming the number of contracts is independent of the number completed, and that contracts arrive at the start of the month, find the distribution of X .

PROBLEM SHEET 6 – PMF AND CDF

Q20

MEDIUM

Which of the following functions satisfy the conditions for a probability mass function?

1. $p(x) = 0.25(0.75)^x$, $x = 0, 1, 2, \dots$ (Hint: geometric series $\sum_{i=0}^{\infty} a^i = 1/(1-a)$ for $|a| < 1$.)
2. $p(x) = \frac{1}{5}(2x - 3)$, $x = 0, 1, 2, 3, 4, 5$
3. $p(x) = \frac{\binom{5}{x}\binom{4}{x-2}}{36}$, $x = 0, 1, 2$

Q21

MEDIUM

Consider the function $p(x) = k/x!$ for $x = 0, 1, 2, 3, 4$, and $p(x) = 0$ otherwise.

1. For what value of k is $p(x)$ a probability mass function?
2. Find $P(X < 2)$.

Q22

MEDIUM

The cumulative distribution function of X is:

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{2} & 0 \leq x < 1 \\ \frac{3}{5} & 1 \leq x < 2 \\ \frac{4}{5} & 2 \leq x < 3 \\ \frac{9}{10} & 3 \leq x < 4 \\ 1 & x \geq 4 \end{cases}$$

Calculate the probability mass function of X .

PROBLEM SHEET 7 – EXPECTATION AND VARIANCE

Q23

MEDIUM

Suppose that a softball player's number of hits X in his next 3 bats satisfies $P(X = 1) = 0.3$, $P(X = 2) = 0.2$, $P(X = 0) = 3P(X = 3)$. Find $E(X)$ and $V(X)$.

Q24

HARD

Use the pool-building company scenario from Q19. Find the expected value and variance of X , the number of contracts carried forward.

Q25

HARD

The density function for a random variable X is:

$$f(x) = \begin{cases} \frac{3}{4}(kx - x^2) & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Determine the value of k , then find $E(X)$ and $V(X)$.

Solutions — Problem Sheets 5, 6 & 7

Full worked solutions for Random Variable questions Q16–Q25.

SOLUTIONS – PROBLEM SHEET 5

A16

1. **Continuous RV.** Time is measured on a continuous scale; range $(0, \infty)$.
2. **Discrete RV.** Word count is a non-negative integer; range $\{0, 1, 2, \dots\}$.
3. **Continuous RV.** Temperature can take any real value in some interval, e.g. $(-10, 50)^\circ\text{C}$.
4. **Discrete RV.** Count of letters sent; range $\{1, 2, 3, \dots\}$.
5. **Continuous RV.** A proportion lies in $[0, 1]$; any value there is possible.

A17

1. Let $M = \max(\text{roll}_1, \text{roll}_2)$. The 36 equally likely outcomes give:

$$P(M = k) = \frac{2k - 1}{36}, \quad k = 1, 2, 3, 4, 5, 6.$$

e.g. $P(M = 3) = 5/36$ (pairs where max is 3: $(1,3), (2,3), (3,3), (3,2), (3,1)$).

2. Let $T = \text{roll}_1 + \text{roll}_2$. T ranges from 2 to 12.

$$P(T = k) = \frac{6 - |k - 7|}{36}, \quad k = 2, 3, \dots, 12.$$

e.g. $P(T = 7) = 6/36 = 1/6$ (maximum), $P(T = 2) = P(T = 12) = 1/36$.

3. Let $D = \text{roll}_1 - \text{roll}_2$. D ranges from -5 to 5 .

$$P(D = k) = \frac{6 - |k|}{36}, \quad k = -5, -4, \dots, 4, 5.$$

The distribution is symmetric about 0.

A18

Total $\binom{10}{2} = 45$ equally likely pairs. Possible winnings:

$X = -20$: both white. $\binom{6}{2} = 15$ ways. $P(X = -20) = 15/45 = 1/3$.

$X = -10 + 20 = 10$: one white, one black. $6 \times 2 = 12$ ways. $P(X = 10) = 12/45 = 4/15$.

$X = 40$: both black. $\binom{2}{2} = 1$ way. $P(X = 40) = 1/45$.

$X = -10$: one white, one red. $6 \times 2 = 12$ ways. $P(X = -10) = 12/45 = 4/15$.

$X = 20$: one black, one red. $2 \times 2 = 4$ ways. $P(X = 20) = 4/45$.

$X = 0$: both red. $\binom{2}{2} = 1$ way. $P(X = 0) = 1/45$.

Check: $15 + 12 + 1 + 12 + 4 + 1 = 45$. ✓

A19

Let C = contracts received this month ($C \in \{1, 2, 3, 4\}$) and K = contracts completed ($K \in \{2, 3\}$, each with prob $1/2$). Total contracts available = $2 + C$. Carried forward = $\max(0, 2 + C - K)$.

Since $K \geq 2$ and available ≥ 3 , carried forward = $2 + C - K \geq 0$ always (minimum is $2 + 1 - 3 = 0$).

$$X = 2 + C - K$$

Range: $X \in \{0, 1, 2, 3, 4, 5\}$. By full enumeration of (C, K) pairs weighted by $P(C) \times \frac{1}{2}$:

X	$P(X = X)$
0	$P(C = 1)P(K = 3) = 0.1 \times 0.5 = 0.05$
1	$P(C = 1, K = 2) + P(C = 2, K = 3) = 0.05 + 0.10 = 0.15$
2	$P(C = 2, K = 2) + P(C = 3, K = 3) = 0.10 + 0.25 = 0.35$
3	$P(C = 3, K = 2) + P(C = 4, K = 3) = 0.25 + 0.10 = 0.35$
4	$P(C = 4, K = 2) = 0.10$

Sum = $0.05 + 0.15 + 0.35 + 0.35 + 0.10 = 1.00$. ✓

A20

1. $\sum_{x=0}^{\infty} 0.25(0.75)^x = 0.25 \cdot \frac{1}{1-0.75} = 0.25 \times 4 = 1$. All values ≥ 0 . **Valid p.m.f.**

2. Check $x = 0$: $p(0) = \frac{1}{5}(0-3) = -0.6 < 0$. **Not a p.m.f.** — negative values violate condition (b).

3. $p(0) = \frac{\binom{5}{0}\binom{4}{-2}}{36}$. But $\binom{4}{-2}$ is undefined (negative lower index). **Not a p.m.f.** as written.

A21

1. Require $\sum_{x=0}^4 k/x! = 1$.

$$\sum = k \left(\frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} \right) = k \left(1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} \right) = k \cdot \frac{65}{24}.$$

So $k = 24/65 \approx 0.3692$.

2. $P(X < 2) = p(0) + p(1) = \frac{24/65}{1} + \frac{24/65}{1} = \frac{48}{65} \approx 0.738$

A22

The p.m.f. is recovered from the jumps in F :

$$p(x) = F(x) - F(x^-)$$

$$p(0) = F(0) - F(0^-) = \frac{1}{2} - 0 = \frac{1}{2}$$

$$p(1) = F(1) - F(0) = \frac{3}{5} - \frac{1}{2} = \frac{1}{10}$$

$$p(2) = F(2) - F(1) = \frac{4}{5} - \frac{3}{5} = \frac{1}{5}$$

$$p(3) = F(3) - F(2) = \frac{9}{10} - \frac{4}{5} = \frac{1}{10}$$

$$p(4) = F(4) - F(3) = 1 - \frac{9}{10} = \frac{1}{10}$$

Check: $1/2 + 1/10 + 1/5 + 1/10 + 1/10 = 10/10 = 1$. ✓

A23

Total probability = 1: $P(X = 0) + 0.3 + 0.2 + P(X = 3) = 1$ and $P(X = 0) = 3P(X = 3)$.

So $3P(X = 3) + P(X = 3) = 0.5 \Rightarrow P(X = 3) = 0.125, P(X = 0) = 0.375$.

$$E(X) = 0(0.375) + 1(0.3) + 2(0.2) + 3(0.125) = 0 + 0.3 + 0.4 + 0.375 =$$

1.075

$$E(X^2) = 0 + 1(0.3) + 4(0.2) + 9(0.125) = 0.3 + 0.8 + 1.125 = 2.225$$

$$V(X) = E(X^2) - [E(X)]^2 = 2.225 - 1.075^2 = 2.225 - 1.1556 =$$

1.069

A24

Using the distribution from A19:

$$\begin{aligned} E(X) &= 0(0.05) + 1(0.15) + 2(0.35) + 3(0.35) + 4(0.10) \\ &= 0 + 0.15 + 0.70 + 1.05 + 0.40 = \end{aligned}$$

2.30

$$E(X^2) = 0 + 0.15 + 4(0.35) + 9(0.35) + 16(0.10) = 0.15 + 1.40 + 3.15 + 1.60 = 6.30$$

$$V(X) = 6.30 - 2.30^2 = 6.30 - 5.29 =$$

1.01

A25

Find k : $\int_0^2 \frac{3}{4}(kx - x^2) dx = 1.$

$$\frac{3}{4} \left[k \frac{x^2}{2} - \frac{x^3}{3} \right]_0^2 = \frac{3}{4} \left(2k - \frac{8}{3} \right) = 1 \Rightarrow 2k - \frac{8}{3} = \frac{4}{3} \Rightarrow k =$$

2

So $f(x) = \frac{3}{4}(2x - x^2) = \frac{3}{2}x - \frac{3}{4}x^2$ on $[0, 2]$.

$$E(X): \int_0^2 x \left(\frac{3}{2}x - \frac{3}{4}x^2 \right) dx = \int_0^2 \left(\frac{3}{2}x^2 - \frac{3}{4}x^3 \right) dx = \left[\frac{1}{2}x^3 - \frac{3}{16}x^4 \right]_0^2 = 4 - 3 = 1$$

$$E(X^2): \int_0^2 x^2 \left(\frac{3}{2}x - \frac{3}{4}x^2 \right) dx = \left[\frac{3}{8}x^4 - \frac{3}{20}x^5 \right]_0^2 = 6 - \frac{96}{20} = 6 - 4.8 = 1.2$$

$$V(X) = 1.2 - 1^2 =$$

0.2

Discrete Probability Distributions

§5.1 - BINOMIAL DISTRIBUTION

Bernoulli Trial & the Binomial Setup

A **Bernoulli trial** is a single experiment with exactly two outcomes: *success* (prob p) and *failure* (prob $1 - p$).

Perform n independent Bernoulli trials. Let X = number of successes. Then $X \sim \text{Bin}(n, p)$ provided:

1. Fixed number of trials n .
2. Each trial is independent.
3. Each trial has the same probability of success p .
4. Only two outcomes per trial.

DEFINITION 5.1.2

Binomial PMF, Mean and Variance

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n$$

$$E(X) = np \quad V(X) = np(1-p)$$

The factor $\binom{n}{x}$ counts the number of sequences with exactly x successes; $p^x(1-p)^{n-x}$ is the probability of any one such sequence (by independence).

COMPLEMENTARY PROBABILITY

At Least / At Most

$$P(X \geq k) = 1 - P(X \leq k-1) = 1 - \sum_{x=0}^{k-1} p(x)$$

Always easier to compute the complement when "at least k " has many terms.

SYMMETRY

Flipping Success/Failure

If $X \sim \text{Bin}(n, p)$ counts successes, then

$n - X \sim \text{Bin}(n, 1-p)$ counts failures.

Useful when $p > 0.5$: compute probabilities for $n - X$ with $q = 1 - p$ instead.

EDGE CASE — WHEN IS BINOMIAL APPROPRIATE?

Sampling with vs without replacement: Strictly, the Binomial requires independence (each trial unaffected by previous ones). Sampling *without* replacement from a finite population violates independence — the correct model is the Hypergeometric distribution. However, if the population is large relative to n (rule of thumb: $n < 5\%$ of population), the Binomial is an acceptable approximation.

EDGE CASE — BOUNDARY VALUES $P=0$ AND $P=1$

If $p = 0$: $P(X = 0) = 1$ and $P(X > 0) = 0$ — all trials fail with certainty. If $p = 1$: $P(X = n) = 1$ — all trials succeed with certainty. Both give $V(X) = 0$. These are degenerate cases; the Binomial formula still works formally.

§5.2 — POISSON DISTRIBUTION

Poisson Process Conditions

The Poisson distribution models the number of events in a fixed interval (time, space, etc.) when:

1. Events occur at random and independently of one another.
2. The average rate λ (events per unit interval) is constant.
3. Two events cannot occur at exactly the same instant (no simultaneous events).

DEFINITION 5.2.1

Poisson PMF, Mean and Variance

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

$$E(X) = \lambda \quad V(X) = \lambda$$

The mean and variance are both equal to λ . This equality is a useful diagnostic: if sample mean $\approx$ sample variance, Poisson may be a good model.

SCALING THE RATE

Changing the Time Interval

If events occur at rate λ per unit time, then over an interval of length t :

$$X \sim \text{Poi}(\lambda t)$$

e.g. 12 breakdowns per 5-day week = 2.4 per day. For 2 days:

$$\lambda t = 4.8.$$

COMPLEMENTARY EVENT

At Least One

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-\lambda}$$

Very commonly needed. When λ is small, $P(X \geq 1) \approx \lambda$.

EDGE CASE – POISSON AS LIMIT OF BINOMIAL

When $n \rightarrow \infty, p \rightarrow 0$, with $np = \lambda$ fixed, $\text{Bin}(n, p) \rightarrow \text{Poi}(\lambda)$. Practical rule: use the Poisson approximation to the Binomial when $n \geq 20$ and $p \leq 0.05$ (or $n \geq 100$ and $np \leq 10$). The approximation improves as $n \rightarrow \infty$ and $p \rightarrow 0$.

EDGE CASE – ADDITIVE PROPERTY

If $X \sim \text{Poi}(\lambda_1)$ and $Y \sim \text{Poi}(\lambda_2)$ are independent, then $X + Y \sim \text{Poi}(\lambda_1 + \lambda_2)$. This is useful when combining two independent Poisson processes (e.g. two separate arrival streams into one queue).

§5.3 – GEOMETRIC DISTRIBUTION

Geometric Distribution

Let X = number of *failures* before the first success in a sequence of independent Bernoulli trials with success probability p . Then $X \sim G(p)$ with:

$$p(x) = p(1-p)^x, \quad x = 0, 1, 2, \dots$$

$$E(X) = \frac{1-p}{p} \quad V(X) = \frac{1-p}{p^2}$$

The total number of trials to first success is $T = X + 1 \sim \text{Geometric}$ with $P(T = t) = p(1-p)^{t-1}$, $t = 1, 2, \dots$

§5.3 – NEGATIVE BINOMIAL DISTRIBUTION

Fix r successes; count X = number of **failures** before the r -th success.

$$p(x) = \binom{x+r-1}{x} p^r (1-p)^x, \quad x = 0, 1, 2, \dots$$

$$E(X) = \frac{r(1-p)}{p} \qquad V(X) = \frac{r(1-p)}{p^2}$$

The Geometric is a special case with $r = 1$.

EDGE CASE – GEOMETRIC MEMORYLESS PROPERTY

The geometric distribution is the **only** discrete distribution with the memoryless property:
 $P(X > m + n \mid X > m) = P(X > n)$. Past failures tell you nothing about when the first success will come. This contrasts with most real-life situations — if you’ve already applied for 10 jobs and got no offers, the geometric model says your next application has exactly the same chance as the first one.

DISTRIBUTION	PMF $P(X)$	$E(X)$	$V(X)$	KEY CONDITION
Bernoulli(p)	$p^x (1-p)^{1-x}, \ x \in \{0, 1\}$	p	$p(1-p)$	Single trial
Bin(n, p)	$\binom{n}{x} p^x (1-p)^{n-x}$	np	$np(1-p)$	n indep. Bernoulli trials
Poi(λ)	$\frac{e^{-\lambda} \lambda^x}{x!}$	λ	λ	Random events in fixed interval
Geometric(p)	$p(1-p)^x, \ x \geq 0$	$\frac{1-p}{p}$	$\frac{1-p}{p^2}$	Failures before 1st success
NegBin(r, p)	$\binom{x+r-1}{x} p^r (1-p)^x$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$	Failures before r -th success

◆ Problem Sheets 8 & 9 — Discrete Distributions

Questions from Problem Sheets 8 and 9 covering Binomial, Poisson, Geometric and Negative Binomial.

PROBLEM SHEET 8 – BINOMIAL AND POISSON

Q26

EASY

A true-false test has five questions. To pass you need at least four correct answers. If you guess each answer randomly, what is the probability that you pass?

Q27

MEDIUM

Airplane engines operate independently and fail with probability 0.1. A plane makes a successful flight if *at most half* its engines fail. Find the probability of a successful flight for:

1. A two-engined plane.
2. A four-engined plane.

Q28

MEDIUM

A right-turn lane at an intersection has capacity for 3 cars. Suppose 30% of cars approaching want to turn right. If a string of 15 cars approaches, what is the probability the lane will be *insufficient* (i.e. more than 3 cars want to turn right)?

Q29

MEDIUM

An ice-cream vendor's sales follow a Poisson distribution averaging 10 sales per hour.

1. What is the probability of selling at least one ice-cream in the first hour?
2. How large must his stock be so that he is at least 95% sure of not running out in any given hour?

Q30

MEDIUM

A pump fails on average once every 500 hours of operation. What is the probability it fails exactly 3 times in 2000 hours of operation?

PROBLEM SHEET 9 – BINOMIAL, POISSON, GEOMETRIC & NEGATIVE BINOMIAL

Q31

HARD

A TV manufacturer receives components in consignments. Quality control: test a random sample of 10; reject if 1 or more are defective.

1. Find the probability of rejection if the true proportion defective is (i) 1%, (ii) 10%, (iii) 30%.
2. If a sample of 20 is tested and the consignment is rejected if 2 or more are defective, find rejection probabilities for the same proportions.
3. Which procedure do you consider better, and why?

Q32

MEDIUM

A factory store has average demand for 8 electric motors per week. Delivery takes one week after ordering. If demand follows a Poisson distribution, how low should the storeman let his stock fall before ordering, if he wants at least 95% certainty of meeting all demand while waiting for delivery?

Q33

MEDIUM

A fair die is tossed until a 6 appears. Find the probability that a 6 has not yet appeared after:

1. The first 10 tosses (i.e. first 6 appears on toss 11 or later).
2. The first 6 appears on exactly the 6th toss.

Q34

HARD

A presidential candidate needs 50 signatures for nomination. Each delegate approached gives a signature with probability 0.15. What is the probability that 300 or more delegates must be approached before collecting the 50th signature?

Q35

MEDIUM

A tuckshop has 10 cartons of milk, of which 3 are sour. You randomly select 4 cartons. What is the probability that at least 2 of the 4 selected are sour? (Use the Hypergeometric distribution.)

Solutions — Problem Sheets 8 & 9

Full worked solutions for Discrete Distribution questions Q26–Q35.

SOLUTIONS – PROBLEM SHEET 8

A26

 $X \sim \text{Bin}(5, 0.5)$. Pass means $X \geq 4$.

$$P(X \geq 4) = P(X = 4) + P(X = 5) = \binom{5}{4}(0.5)^5 + \binom{5}{5}(0.5)^5 = \frac{5+1}{32} =$$

$$\frac{6}{32} = 0.1875$$

A27

1. **Two engines:** $X \sim \text{Bin}(2, 0.1)$. Successful if $X \leq 1$ (at most 1 failure = at most half).

$$P(X \leq 1) = P(X = 0) + P(X = 1) = 0.9^2 + 2(0.1)(0.9) = 0.81 + 0.18 =$$

$$0.99$$

2. **Four engines:** $X \sim \text{Bin}(4, 0.1)$. Successful if $X \leq 2$.

$$\begin{aligned} P(X \leq 2) &= \sum_{x=0}^2 \binom{4}{x} (0.1)^x (0.9)^{4-x} \\ &= 0.9^4 + 4(0.1)(0.9)^3 + 6(0.01)(0.81) = 0.6561 + 0.2916 + 0.0486 = \end{aligned}$$

$$0.9963$$

The four-engined plane is more reliable.

A28

Let $X \sim \text{Bin}(15, 0.3)$ = number wanting to turn right. Lane insufficient when $X > 3$.

$$P(X > 3) = 1 - P(X \leq 3) = 1 - \sum_{x=0}^3 \binom{15}{x} (0.3)^x (0.7)^{15-x}$$

$$P(X = 0) = (0.7)^{15} = 0.00475$$

$$P(X = 1) = 15(0.3)(0.7)^{14} = 0.03052$$

$$P(X = 2) = 105(0.09)(0.7)^{13} = 0.09156$$

$$P(X = 3) = 455(0.027)(0.7)^{12} = 0.17004$$

$$P(X \leq 3) = 0.00475 + 0.03052 + 0.09156 + 0.17004 = 0.29687$$

$$P(X > 3) = 1 - 0.297 \approx$$

0.703

A29

1. $X \sim \text{Poi}(10)$.

$$P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-10} = 1 - 0.0000454 \approx$$

0.99995

2. Need smallest k such that $P(X \leq k) \geq 0.95$. Using the Poisson CDF with $\lambda = 10$:

$$P(X \leq 14) \approx 0.9165; P(X \leq 15) \approx 0.9513.$$

So he needs a stock of at least **15 ice-creams**.

A30

Average rate: 1 failure per 500 hours = $\lambda = 1/500$ per hour. Over 2000 hours: $\lambda t = 2000/500 = 4$.

$X \sim \text{Poi}(4)$.

$$P(X = 3) = \frac{e^{-4} \cdot 4^3}{3!} = \frac{e^{-4} \cdot 64}{6} = \frac{0.01832 \times 64}{6} \approx$$

0.1954

A31

1. Procedure 1: $X \sim \text{Bin}(10, p)$. Reject if $X \geq 1$, so $P(\text{reject}) = 1 - (1 - p)^{10}$.

(i) $p = 0.01$: $P = 1 - (0.99)^{10} = 1 - 0.9044 = 0.096$

(ii) $p = 0.10$: $P = 1 - (0.90)^{10} = 1 - 0.3487 = 0.651$

(iii) $p = 0.30$: $P = 1 - (0.70)^{10} = 1 - 0.0282 = 0.972$

2. Procedure 2: $X \sim \text{Bin}(20, p)$. Reject if $X \geq 2$, so $P(\text{reject}) = 1 - P(X = 0) - P(X = 1)$.

(i) $p = 0.01$: $P = 1 - (0.99)^{20} - 20(0.01)(0.99)^{19} = 1 - 0.8179 - 0.1652 = 0.017$

(ii) $p = 0.10$: $P = 1 - (0.90)^{20} - 20(0.1)(0.9)^{19} = 1 - 0.1216 - 0.2702 = 0.608$

(iii) $p = 0.30$: $P = 1 - (0.70)^{20} - 20(0.3)(0.7)^{19} = 1 - 0.0008 - 0.0068 = 0.992$

3. Procedure 1 is better at detecting low-defect batches ($p = 1\%$: 9.6% vs 1.7% rejection rate). Procedure 2 is marginally better for high-defect batches. For consumer protection against borderline batches ($p = 10\%$),

Procedure 1 (65.1%) significantly outperforms Procedure 2 (60.8%). **Procedure 1 is generally preferred.**

A32

$X \sim \text{Poi}(8)$. Need smallest stock level k such that $P(X \leq k) \geq 0.95$.

Cumulate: $P(X \leq 9) \approx 0.7166$; $P(X \leq 10) \approx 0.8159$; $P(X \leq 11) \approx 0.8881$; $P(X \leq 12) \approx 0.9362$; $P(X \leq 13) \approx 0.9658$.

$P(X \leq 12) = 0.9362 < 0.95$ but $P(X \leq 13) = 0.9658 \geq 0.95$.

The storeman should order when stock falls to **13 motors**.

A33

1. $X \sim G(p = 1/6)$ counts failures (non-6 rolls) before the first 6. "No 6 in first 10 tosses" means $X \geq 10$.

$$P(X \geq 10) = (1 - 1/6)^{10} = (5/6)^{10} \approx$$

0.1615

2. "First 6 on exactly the 6th toss" means 5 failures then 1 success: $P(X = 5) = \frac{1}{6}(5/6)^5 \approx 0.0670$.

Alternatively using the NegBin with $r = 6$ — see Q34 for that approach.

A34

$X \sim \text{NegBin}(r = 50, p = 0.15)$ counts failures before the 50th success.

"300 or more delegates approached before 50th signature" means the 50th signature is on delegate 300 or later, i.e. there are at least $300 - 50 = 250$ failures before the 50th success: $P(X \geq 250)$.

Since $r = 50$, $E(X) = \frac{50(0.85)}{0.15} = \frac{42.5}{0.15} \approx 283.3$ and $V(X) = \frac{50(0.85)}{0.15^2} \approx 1888.9$, giving $\text{SD} \approx 43.5$.

For large r , approximate via Normal: standardise $z = \frac{250 - 283.3}{43.5} \approx -0.765$.

$$P(X \geq 250) \approx P(Z \geq -0.765) = \Phi(0.765) \approx$$

0.778

A35

Hypergeometric model: population $N = 10$, defectives (sour) $K = 3$, good $N - K = 7$, sample $n = 4$.

$$P(X = k) = \frac{\binom{3}{k} \binom{7}{4-k}}{\binom{10}{4}}, \quad k = 0, 1, 2, 3$$

$$\binom{10}{4} = 210.$$

$$P(X = 0) = \frac{\binom{3}{0} \binom{7}{4}}{210} = \frac{1 \times 35}{210} = \frac{35}{210}$$

$$P(X = 1) = \frac{\binom{3}{1} \binom{7}{3}}{210} = \frac{3 \times 35}{210} = \frac{105}{210}$$

$$P(X = 2) = \frac{\binom{3}{2} \binom{7}{2}}{210} = \frac{3 \times 21}{210} = \frac{63}{210}$$

$$P(X = 3) = \frac{\binom{3}{3} \binom{7}{1}}{210} = \frac{1 \times 7}{210} = \frac{7}{210}$$

$$P(X \geq 2) = P(X = 2) + P(X = 3) = \frac{63 + 7}{210} = \frac{70}{210} =$$

$$\frac{1}{3} \approx 0.333$$